24 Lara is a skier.

She completed a ski race in 1 minute 54 seconds.
The race was 475 m in length.

Lara assumes that her average speed is the same for each race.

(a) Using this assumption, work out how long Lara should take to complete a 700 m race.
Give your answer in minutes and seconds.

1 minute 54 seconds = 114 seconds

	Metres	Seconds	
	475	114	
÷475	1	0.24	÷475
×700	700	168	×700

1 mark
Scale the time to one metre

1 mark
Scale to the 700 m race

168 seconds = 2 × 60 seconds + 48 seconds

1 mark
2 minutes 48 seconds

2 minutes 48 seconds

(3)

Lara's average speed actually increases the further she goes.

(b) How does this affect your answer to part (a)?

She will take less than 2 minutes 48 seconds.

My estimate in part (a) is too long.

1 mark
Greater average speed means less time for 700 m

(1)

(Total for Question 24 is 4 marks)